

From DK1 Data to Forecasts, Decisions, and Settlement Traces: Midday–Evening Coupling in Energy and Upward aFRR

Abstract

This document states the empirical claim supported by the present DK1 study and connects the complete analysis chain: observed market and power-system data, gate-specific probabilistic forecasts, joint day-ahead energy and upward-aFRR decisions, and ex-post 15-minute settlement traces. Actual renewable curtailment is not observed. We therefore distinguish true renewable utilization from a reproducible VRE-above-load proxy. In 330 complete DK1 days, the proxy is positive in 2,097 of 7,920 hours (26.48%). Nevertheless, all 2,097 are net-export hours: 43.77% of export energy is supplied by non-VRE generation, while Power-to-Heat has already absorbed 679 GWh. A gate-safe weather–load model predicts proxy-positive hours with 79.19% precision and 61.92% recall at 07:30; the 12:00 update raises these to 80.66% and 66.10%. The evaluated schedules still overlap with only 0.0116–0.1408% of the midday proxy energy. Evening upward-aFRR commitments use 88.2–98.1% of the battery’s available power-duration envelope and deliver 95.64–96.82% of requested evening activation. Physical midday-to-evening coupling therefore exists through state of charge, and a usable but conservative gate-time proxy signal exists, but the current price-driven optimizer does not consume that signal.

1 Scope and terminology

This section distinguishes system-level renewable energy utilization from ex-post proxy metrics that can be evaluated using the available DK1 data and simulated battery schedules. The repository-wide terminology and evidence rules are defined in `CONVENTIONS.md`. Variable renewable energy (VRE) refers only to wind and solar generation. The DK1 data do not contain unconstrained available VRE generation, curtailment instructions, observed curtailed energy, or source tracing for battery charging. Consequently, the DK1 proxy metrics defined below must not be reported as actual curtailment reduction or as an observed renewable utilization rate.

Let $t \in \mathcal{T}$ denote a 15-minute settlement interval and let $\Delta t = 0.25$ h. UTC timestamps remain the unique data keys and are mapped to Europe/Copenhagen local delivery time only when defining reporting windows. For delivery day d , the midday and evening-peak sets are fixed as

$$\mathcal{M}_d = \{t : t \text{ falls between 10:00 and 15:59 on day } d\}, \quad (1)$$

$$\mathcal{E}_d = \{t : t \text{ falls between 17:00 and 21:59 on day } d\}. \quad (2)$$

All UTC observations are retained on daylight-saving transition days; repeated local clock times are not deduplicated.

2 Data-to-settlement research chain

For delivery day d , the reproducible analysis is organized as the following ordered map:

$$\mathcal{D}_{<d} \xrightarrow{\text{forecast at gate } g} \hat{\Xi}_{d,s}^{(g)} \xrightarrow{\text{stochastic decision}} u_d \xrightarrow{\text{realized DK1 observations}} z_{d,t} \xrightarrow{\text{diagnostics}} J_d. \quad (3)$$

Here, $\mathcal{D}_{<d}$ contains only observations available before delivery day d ; $g \in \{07:30, 12:00\}$ is the information gate; $s \in \mathcal{S}_d$ indexes a forecast scenario; u_d denotes the deployed energy and reserve decisions; $z_{d,t}$ is the realized 15-minute physical and financial ledger; and J_d is the vector of midday–evening diagnostic outcomes. Ex-post wind, solar, and load observations enter only the final diagnostic stage and never enter the corresponding day-ahead decision.

The forecasting target vector for scenario s is

$$\hat{\Xi}_{d,s}^{(g)} = \left\{ \hat{\lambda}_{d,s,t}^{\text{DA},(g)}, \hat{\kappa}_{d,s,h}^{\text{up},(g)}, \hat{a}_{d,s,t}^{(g)}, \hat{\lambda}_{d,s,t}^{\text{act},(g)}, \hat{\lambda}_{d,s,t}^{\text{imb},(g)} \right\}, \quad (4)$$

where λ^{DA} is the day-ahead energy price, κ^{up} is the upward-aFRR capacity price, a is the activation-energy coefficient, λ^{act} is the activation price, and λ^{imb} is the imbalance price used for shortfall and recovery settlement. The experiment uses 50 equal-weight scenarios. The 07:30 solve chooses the hourly reserve vector. At 12:00 the realized capacity price and updated energy-side information are introduced, while the 07:30 reserve commitment remains frozen. The 12:00 programme first maximizes expected activation delivery and then maximizes the risk-adjusted profit within a 10^{-9} MWh delivery tolerance.

The observed DK1 inputs and their native resolutions are

$$\mathcal{D}^{\text{market}} = \{\lambda_t^{\text{DA}}, \kappa_h^{\text{up}}, Q_h^{\text{proc}}, A_t^{\text{up}}, \lambda_t^{\text{act}}, \lambda_t^{\text{imb}}\}, \quad (5)$$

$$\mathcal{D}^{\text{proxy}} = \{P_h^{\text{wind,obs}}, P_h^{\text{solar,obs}}, L_h^{\text{obs}}, F_{b,h}, C_{k,h}, H_h\}. \quad (6)$$

Here, $F_{b,h}$ is the signed exchange on border b (positive for imports), $C_{k,h}$ is generation in non-VRE category k , and H_h is observed Power-to-Heat consumption. Gate-specific ECMWF weather runs are introduced below as forecast inputs rather than as ex-post observations. Energy, activation, and imbalance settlement use 15-minute intervals; aFRR capacity and the proxy inputs use hourly observations. Hourly quantities are mapped to their four settlement intervals without changing the UTC key.

3 System-level renewable energy utilization

For VRE technology $v \in \mathcal{V} = \{\text{wind}, \text{solar}\}$, let $P_{v,t}^{\text{avail}}$ denote the power that would be available without curtailment, and let $P_{v,t}^{\text{used}}$ denote the power accepted by the system after dispatch. The actual curtailed power is

$$P_{v,t}^{\text{curt}} = \left[P_{v,t}^{\text{avail}} - P_{v,t}^{\text{used}} \right]^+, \quad [x]^+ = \max(x, 0). \quad (7)$$

For an evaluation horizon $\mathcal{H} \subseteq \mathcal{T}$, curtailed energy, the curtailment rate, and the renewable utilization rate are

$$E_{\mathcal{H}}^{\text{curt}} = \sum_{t \in \mathcal{H}} \sum_{v \in \mathcal{V}} P_{v,t}^{\text{curt}} \Delta t, \quad (8)$$

$$CR_{\mathcal{H}} = \frac{E_{\mathcal{H}}^{\text{curt}}}{\sum_{t \in \mathcal{H}} \sum_{v \in \mathcal{V}} P_{v,t}^{\text{avail}} \Delta t}, \quad (9)$$

$$UR_{\mathcal{H}} = 1 - CR_{\mathcal{H}}. \quad (10)$$

Equations (7)–(10) require available and utilized generation to share the same technology, location, and temporal boundaries. Realized wind and solar generation already reflect dispatch, network constraints, and possible curtailment; they therefore cannot substitute for $P_{v,t}^{\text{avail}}$.

The contribution of battery storage to curtailment reduction requires a controlled system-level counterfactual. Let scenario 0 exclude the battery and scenario 1 include it, while holding

demand, weather, available generation, network constraints, and all other generating-unit constraints fixed. The model-based curtailment reduction attributable to the battery is

$$\Delta E_{\text{BESS},\mathcal{H}}^{\text{curt}} = E_{\mathcal{H}}^{\text{curt},0} - E_{\mathcal{H}}^{\text{curt},1}. \quad (11)$$

A positive value in Eq. (11) can be reported as model-based curtailment reduction. A claim that the battery is charged from VRE additionally requires source tracing or an explicit coupling constraint, for example

$$0 \leq P_t^{\text{ch,VRE}} \leq \sum_{v \in \mathcal{V}} P_{v,t}^{\text{used}}, \quad P_t^{\text{ch,VRE}} \leq P_t^{\text{ch}}. \quad (12)$$

The current price-taking DK1 simulation does not provide the source tracing required by Eq. (12); hence, $P_t^{\text{ch,VRE}}$ is not evaluated.

4 DK1 observations and model-generated variables

Table 1: Variables used in the utilization definitions and DK1 proxy metrics.

Symbol	Data role	Definition
$P_t^{\text{wind,obs}}$	DK1 ex-post observation	Realized wind generation
$P_t^{\text{solar,obs}}$	DK1 ex-post observation	Realized solar generation
L_t^{obs}	DK1 ex-post observation	Total load
$F_{b,t}$	DK1 ex-post observation	Signed exchange on border b ; import positive
$C_{k,t}$	DK1 ex-post observation	Generation in non-VRE category k
H_t	DK1 ex-post observation	Power-to-Heat consumption included in total load
$w_{d,h}^{(g)}$	Archived forecast	Weather vector visible at gate g
$\hat{L}_{d,h}^{(g)}$	Model-generated	Gate-specific load forecast
λ_t^{DA}	DK1 market observation	Day-ahead energy price
κ_h^{up}	DK1 market observation	Upward-aFRR capacity price
a_t	DK1 settlement observation	Required activation coefficient
λ_t^{act}	DK1 settlement observation	Upward activation price
λ_t^{imb}	DK1 settlement observation	Imbalance and recovery price
P_t^{ch}	Model-generated	Battery charging power
P_t^{dis}	Model-generated	Battery discharging power
r_h^{up}	Model-generated	Hourly upward-aFRR commitment
e_t^{del}	Model-generated	Delivered activation energy
P_t^{rec}	Model-generated	Paid SOC-recovery power
S_t	Model-generated	Battery state of charge
$P_{v,t}^{\text{avail}}$	Unobserved	Unconstrained VRE availability
$P_{v,t}^{\text{curt}}$	Unobserved	Actual curtailed VRE power

Because of publication delays, $P_t^{\text{wind,obs}}$, $P_t^{\text{solar,obs}}$, and L_t^{obs} are used only for ex-post evaluation and are not inputs to the corresponding day-ahead decisions. Battery charging, discharging, and state of charge are outputs of the research model rather than observations from a physical DK1 battery.

5 Ex-post proxy metrics observable in DK1

Observed VRE power, the VRE-to-load ratio, the local residual-load proxy, and the local VRE-above-load proxy are defined as

$$P_t^{\text{VRE,obs}} = P_t^{\text{wind,obs}} + P_t^{\text{solar,obs}}, \quad (13)$$

$$\rho_t^{\text{VRE}} = \frac{P_t^{\text{VRE,obs}}}{L_t^{\text{obs}}}, \quad L_t^{\text{obs}} > 0, \quad (14)$$

$$NL_t^{\text{proxy}} = L_t^{\text{obs}} - P_t^{\text{VRE,obs}}, \quad (15)$$

$$X_t^{\text{proxy}} = [-NL_t^{\text{proxy}}]^+ = [P_t^{\text{VRE,obs}} - L_t^{\text{obs}}]^+. \quad (16)$$

Neither NL_t^{proxy} nor X_t^{proxy} accounts for conventional generation, cross-border exchanges, network constraints, or actual curtailment. They must therefore be described as a local residual-load proxy and a local VRE-above-load proxy, respectively, rather than as system net load or actual renewable surplus.

5.1 Exchange, conventional-generation, and Power-to-Heat decomposition

The signed net exchange, gross export, gross import, and observed non-VRE generation are

$$F_t = \sum_{b \in \mathcal{B}} F_{b,t}, \quad E_t = [-F_t]^+, \quad I_t = [F_t]^+, \quad (17)$$

$$C_t = \sum_{k \in \mathcal{K}} C_{k,t}, \quad \mathcal{K} = \{\text{central, local, commercial, local self-consumption, hydro, unknown}\}. \quad (18)$$

The Energinet settlement table obeys the hourly balance

$$P_t^{\text{VRE,obs}} + C_t + F_t = L_t^{\text{obs}} + \varepsilon_t, \quad (19)$$

where ε_t is the numerical balance residual. Substitution into Eq. (16) gives the identity

$$X_t^{\text{proxy}} = [E_t - I_t - C_t + \varepsilon_t]^+. \quad (20)$$

Thus a proxy-positive hour may be an exporting hour rather than an hour in which energy lacks a system use. The export coverage and the conventional share of export over a set $\mathcal{H}$ are reported without adding their numerators:

$$EC_{\mathcal{H}} = \frac{\sum_{t \in \mathcal{H}} \min(X_t^{\text{proxy}}, E_t)}{\sum_{t \in \mathcal{H}} X_t^{\text{proxy}}}, \quad (21)$$

$$CS_{\mathcal{H}} = \frac{\sum_{t \in \mathcal{H}} C_t}{\sum_{t \in \mathcal{H}} E_t}. \quad (22)$$

Power-to-Heat is already contained in L_t^{obs} . It is therefore not subtracted again in Eq. (20). Its observed absorption is evaluated through the explicit no-Power-to-Heat counterfactual

$$X_t^{\text{noP2H}} = [P_t^{\text{VRE,obs}} - (L_t^{\text{obs}} - H_t)]^+, \quad (23)$$

$$A_t^{\text{P2H}} = X_t^{\text{noP2H}} - X_t^{\text{proxy}}, \quad (24)$$

$$HS_{\mathcal{H}} = \frac{\sum_{t \in \mathcal{H}} A_t^{\text{P2H}}}{\sum_{t \in \mathcal{H}} X_t^{\text{noP2H}}}. \quad (25)$$

A_t^{P2H} is an observed accounting counterfactual, not an estimate of curtailment avoided by electric heating.

The simulated midday charging energy on delivery day d is

$$E_{\text{ch},d}^{\text{mid}} = \sum_{t \in \mathcal{M}_d} P_t^{\text{ch}} \Delta t. \quad (26)$$

The temporal overlap between simulated charging and the local VRE-above-load proxy is

$$E_{\text{ch},d}^{\text{overlap}} = \sum_{t \in \mathcal{M}_d} \min(P_t^{\text{ch}}, X_t^{\text{proxy}}) \Delta t. \quad (27)$$

Two proxy ratios with distinct denominators are then reported:

$$AS_d^{\text{ch}} = \frac{E_{\text{ch},d}^{\text{overlap}}}{E_{\text{ch},d}^{\text{mid}}}, \quad E_{\text{ch},d}^{\text{mid}} > 0, \quad (28)$$

$$AR_d^{\text{proxy}} = \frac{E_{\text{ch},d}^{\text{overlap}}}{\sum_{t \in \mathcal{M}_d} X_t^{\text{proxy}} \Delta t}, \quad \sum_{t \in \mathcal{M}_d} X_t^{\text{proxy}} \Delta t > 0. \quad (29)$$

AS_d^{ch} is the share of simulated midday charging that coincides with the local VRE-above-load proxy. AR_d^{proxy} is the share of that proxy whose power magnitude temporally overlaps with simulated charging. Both are temporal alignment metrics. Neither traces the physical origin of electricity nor equals the renewable utilization rate in Eq. (10).

To avoid relying exclusively on the strict condition $P_t^{\text{VRE,obs}} > L_t^{\text{obs}}$, a renewable-rich indicator may be defined using a threshold τ . The threshold must be fixed using the training period or an independent reference period and must not be selected from test outcomes:

$$I_t^{\text{rich}} = \mathbb{I}(\rho_t^{\text{VRE}} \geq \tau), \quad (30)$$

$$E_{\text{ch},d}^{\text{rich}} = \sum_{t \in \mathcal{M}_d} I_t^{\text{rich}} P_t^{\text{ch}} \Delta t. \quad (31)$$

Until τ is frozen on an independent sample, Eq. (31) is a sensitivity metric rather than a confirmatory primary outcome.

Charging during negative day-ahead prices is defined as

$$E_{\text{ch},d}^{\text{neg}} = \sum_{t \in \mathcal{M}_d} \mathbb{I}(\lambda_t^{\text{DA}} < 0) P_t^{\text{ch}} \Delta t. \quad (32)$$

Equation (32) measures an economic response to negative prices. Negative prices can indicate insufficient system flexibility, but they do not independently establish that wind or solar generation was curtailed.

6 Gate-available weather, load, and proxy forecasts

For each delivery day d , archived ECMWF IFS forecasts are selected by model initialization time rather than by the data-download time. Let $r(d, g)$ denote the selected run and impose a conservative six-hour production delay:

$$r(d, 07:30) = (d - 2) \text{ 18:00 UTC}, \quad a(d, 07:30) = r(d, 07:30) + 6 \text{ h}, \quad (33)$$

$$r(d, 12:00) = (d - 1) \text{ 00:00 UTC}, \quad a(d, 12:00) = r(d, 12:00) + 6 \text{ h}. \quad (34)$$

The implementation asserts $a(d, g) \leq G(d, g)$, where $G(d, g)$ is the UTC instant corresponding to the civil-time gate on day $d - 1$. The weather vector $w_{d,h}^{(g)}$ contains the equal-grid DK1 means of 2-m temperature, 100-m wind speed, shortwave radiation, and total cloud cover, together with the spatial standard deviation of 100-m wind speed, from run $r(d, g)$. The run archive and its distinction between initialization time and public availability are documented by Open-Meteo [4].

Weather fields are sampled on a 3×3 grid defined by latitudes $\{55.5, 56.25, 57.0\}^\circ\text{N}$ and longitudes $\{8.0, 9.0, 10.0\}^\circ\text{E}$, yielding nine points over Jutland and adjacent waters. For each valid hour and model run, the weather variables are aggregated using an unweighted arithmetic mean across the nine points; the population standard deviation of 100-m wind speed is additionally retained. No bidding-zone boundary mask, land-sea mask, area weighting, or installed-capacity weighting is applied. These fields should therefore be interpreted as regional DK1 weather proxies rather than exact spatial averages over the DK1 bidding zone.

Separate component models are fitted for $q \in \{\text{wind, solar, load}\}$. The design vector is

$$\phi_{q,d,h}^{(g)} = \left[1, \text{Fourier}_4(h), \text{DoW}(d), w_{d,h}^{(g)}, T_{d,h}^2, U_{d,h}^2, U_{d,h}^3, q_{d-7,h}^{\text{obs}}, q_{d-14,h}^{\text{obs}} \right]^\top, \quad (35)$$

where T and U are temperature and 100-m wind speed, respectively; $\text{Fourier}_4(h)$ contains sine and cosine terms for harmonics one through four; and $\text{DoW}(d)$ contains six weekday indicators with Monday omitted. Ordinary least squares is fitted only on the existing 84-calendar-day training window from $d - 119$ through $d - 36$:

$$\hat{\beta}_{q,d}^{(g)} = \arg \min_{\beta} \sum_{(d',h) \in \mathcal{F}_d} \left(q_{d',h}^{\text{obs}} - \phi_{q,d',h}^{(g)\top} \beta \right)^2, \quad (36)$$

$$\hat{q}_{d,h}^{(g)} = \left[\phi_{q,d,h}^{(g)\top} \hat{\beta}_{q,d}^{(g)} \right]^+. \quad (37)$$

No target from delivery day d enters $\mathcal{F}_d$. Missing inputs remain missing and the affected forecast is not evaluated. The observed 7- and 14-day lags are assigned the same 75.75-hour publication delay used by the DK1 data audit, and the implementation separately verifies their availability at each gate. The gate-available proxy forecast is then

$$\hat{X}_{d,h}^{\text{proxy},g} = \left[\hat{P}_{d,h}^{\text{wind},g} + \hat{P}_{d,h}^{\text{solar},g} - \hat{L}_{d,h}^g \right]^+. \quad (38)$$

Equation (38) is a model-generated forecast of the same ex-post proxy in Eq. (16); it is neither an official TSO load forecast nor a forecast of physical curtailment. It is evaluated in parallel with the frozen storage experiment and is not supplied to the current optimizer; this preserves the existing decision comparison while measuring whether a future proxy-aware policy would have usable gate-time information.

For every gate and evaluation set $\mathcal{Q}$, the implementation reports component MAE, proxy MAE and RMSE, and positive-hour precision and recall:

$$\text{MAE}_q^g = \frac{1}{|\mathcal{Q}|} \sum_{(d,h) \in \mathcal{Q}} \left| \hat{q}_{d,h}^g - q_{d,h}^{\text{obs}} \right|, \quad (39)$$

$$\text{RMSE}_X^g = \sqrt{\frac{1}{|\mathcal{Q}|} \sum_{(d,h) \in \mathcal{Q}} \left(\hat{X}_{d,h}^{\text{proxy},g} - X_{d,h}^{\text{proxy}} \right)^2}, \quad (40)$$

$$\text{Precision}_X^g = \frac{\sum \hat{I}_{d,h}^g I_{d,h}}{\sum \hat{I}_{d,h}^g}, \quad \text{Recall}_X^g = \frac{\sum \hat{I}_{d,h}^g I_{d,h}}{\sum I_{d,h}}, \quad (41)$$

where $I_{d,h} = \mathbb{I}(X_{d,h}^{\text{proxy}} > 0)$ and $\hat{I}_{d,h}^g = \mathbb{I}(\hat{X}_{d,h}^{\text{proxy},g} > 0)$. The information value between the gates is summarized by

$$\text{MAR}_{07:30 \rightarrow 12:00} = \frac{1}{|\mathcal{Q}|} \sum_{(d,h) \in \mathcal{Q}} \left| \hat{X}_{d,h}^{\text{proxy},12:00} - \hat{X}_{d,h}^{\text{proxy},07:30} \right|. \quad (42)$$

7 Forecast-driven energy and upward-aFRR decision

The common battery has rated power $\bar{P} = 4$ MW, energy capacity $\bar{S} = 8$ MWh, round-trip efficiency $\eta^{\text{rt}} = 0.85$, one-way efficiencies $\eta_c = \eta_d = \sqrt{0.85}$, and initial and terminal SOC $S_0 = S_T = 4.4$ MWh. The deployed decision is

$$u_d = \{P_{d,t}^{\text{ch}}, P_{d,t}^{\text{dis}}, r_{d,h}^{\text{up}}, S_{d,t}^{\text{plan}}\}. \quad (43)$$

For scenario s , required upward activation and the realized SOC recursion are

$$e_{d,s,t}^{\text{req}} = r_{d,h(t)}^{\text{up}} a_{d,s,t}, \quad (44)$$

$$S_{d,s,t} = S_{d,s,t-1} + \eta_c \Delta t \left(P_{d,t}^{\text{ch}} + P_{d,s,t}^{\text{rec}} \right) - \frac{\Delta t}{\eta_d} P_{d,t}^{\text{dis}} - \frac{e_{d,s,t}^{\text{del}}}{\eta_d}. \quad (45)$$

The principal physical constraints are

$$0 \leq S_{d,t}^{\text{plan}}, S_{d,s,t} \leq \bar{S}, \quad S_{d,s,T} = S_0, \quad (46)$$

$$P_{d,t}^{\text{ch}} + P_{d,t}^{\text{dis}} + r_{d,h(t)}^{\text{up}} \leq \bar{P}, \quad (47)$$

$$P_{d,t}^{\text{ch}} + P_{d,t}^{\text{dis}} + P_{d,s,t}^{\text{rec}} + \frac{e_{d,s,t}^{\text{del}}}{\Delta t} \leq \bar{P}, \quad (48)$$

$$S_{d,t}^{\text{plan}} \geq \frac{T^{\text{res}}}{\eta_d} r_{d,h(t)}^{\text{up}}, \quad T^{\text{res}} = 1 \text{ h}, \quad (49)$$

$$0 \leq e_{d,s,t}^{\text{del}} \leq e_{d,s,t}^{\text{req}}. \quad (50)$$

Equation (47) reserves gross battery power before activation. Equation (48) reveals a different source of shortfall: even when sufficient energy is stored, scheduled energy operation can consume the power headroom required for activation.

The scenario settlement profit is

$$\begin{aligned} \Pi_{d,s}(u_d) = & \sum_t \Delta t \lambda_{d,s,t}^{\text{DA}} \left(P_{d,t}^{\text{dis}} - P_{d,t}^{\text{ch}} \right) + \sum_h \kappa_{d,s,h}^{\text{up}} r_{d,h}^{\text{up}} \\ & + \sum_t \lambda_{d,s,t}^{\text{act}} e_{d,s,t}^{\text{del}} - \sum_t \lambda_{d,s,t}^{\text{imb}} \left(e_{d,s,t}^{\text{req}} - e_{d,s,t}^{\text{del}} \right) \\ & - \sum_t \Delta t \lambda_{d,s,t}^{\text{imb}} P_{d,s,t}^{\text{rec}}. \end{aligned} \quad (51)$$

With scenario probabilities π_s , risk weight χ , and lower-tail confidence level $\beta = 0.95$, the stochastic objective is

$$\max_{u_d} (1 - \chi) \sum_{s \in \mathcal{S}_d} \pi_s \Pi_{d,s}(u_d) + \chi \text{CVaR}_{\beta}^{\text{lower}}(\Pi_{d,s}(u_d)). \quad (52)$$

The 12:00 delivery-first implementation is lexicographic. It first obtains

$$D_d^* = \max_s \sum_s \pi_s \sum_t e_{d,s,t}^{\text{del}}, \quad (53)$$

and then solves Eq. (52) subject to expected delivery of at least $D_d^* - \varepsilon_D$, where $\varepsilon_D = 10^{-9}$ MWh. This prevents the noon update from exchanging material delivery capability for a small profit improvement.

8 Fifteen-minute settlement trace and linkage diagnostics

For each deployed strategy, the ex-post trace records

$$z_{d,t} = \left(P_t^{\text{ch}}, P_t^{\text{dis}}, r_{h(t)}^{\text{up}}, e_t^{\text{req}}, e_t^{\text{del}}, P_t^{\text{rec}}, S_t^{\text{plan}}, S_t^{\text{real}}, \Pi_t^{\text{energy}}, \Pi_t^{\text{cap}}, \Pi_t^{\text{act}}, C_t^{\text{short}}, C_t^{\text{rec}} \right). \quad (54)$$

The ledger is an output layer only: it does not re-optimize or alter u_d . Hourly capacity revenue is allocated equally across the four 15-minute slots, and every trace is required to reproduce the existing daily settlement ledger.

The daily diagnostic vector is

$$J_d = \left(E_{\text{ch},d}^{\text{mid}}, E_{\text{ch},d}^{\text{overlap}}, AS_d^{\text{ch}}, AR_d^{\text{proxy}}, S_{d,17}, R_d^{\text{eve}}, E_d^{\text{req,eve}}, E_d^{\text{del,eve}}, U_d^{\text{eve}}, C_d^{\text{rec}}, \Pi_d \right), \quad (55)$$

where

$$R_d^{\text{eve}} = \sum_{t \in \mathcal{E}_d} r_{h(t)}^{\text{up}} \Delta t, \quad (56)$$

$$E_d^{\text{req,eve}} = \sum_{t \in \mathcal{E}_d} e_t^{\text{req}}, \quad E_d^{\text{del,eve}} = \sum_{t \in \mathcal{E}_d} e_t^{\text{del}}, \quad (57)$$

$$U_d^{\text{eve}} = E_d^{\text{req,eve}} - E_d^{\text{del,eve}}, \quad DR^{\text{eve}} = \frac{\sum_d E_d^{\text{del,eve}}}{\sum_d E_d^{\text{req,eve}}}, \quad (58)$$

$$C_d^{\text{rec}} = \sum_t \Delta t \lambda_t^{\text{imb}} P_t^{\text{rec}}. \quad (59)$$

The observable linkage is summarized by

$$\Gamma_S = \text{Corr} \left(E_{\text{ch},d}^{\text{mid}}, S_{d,17} \right), \quad (60)$$

$$\Gamma_R = \text{Corr} \left(E_{\text{ch},d}^{\text{mid}}, R_d^{\text{eve}} \right). \quad (61)$$

A physically useful midday–evening link requires $\Gamma_S > 0$. A decision protocol that actively converts midday charging into additional evening reserve would also be expected to have $\Gamma_R \geq 0$, unless the reserve decision is already power-saturated or frozen before the midday information becomes available.

9 Empirical evidence from the completed DK1 run

9.1 Data coverage and proxy occurrence

The proxy data cover 1 October 2025 through 27 August 2026. There are 330 complete delivery days and 7,920 valid hourly observations. The missing day 25 August 2026 and the incomplete 28 August 2026 observation are excluded from the proxy statistics. The decision–trace analysis uses the 210 complete days that overlap the out-of-sample decision period.

The full-sample proxy energy is 1,580,970 MWh, of which 427,646 MWh occurs in the fixed midday window. Table 2 shows that the phenomenon is seasonal but not confined to solar noon. DK1 wind generation creates a broad surplus-proxy pattern, with the highest clock-hour frequency at 16:00–17:00. The empirical object is therefore a renewable-rich midday-to-evening transition, not a purely solar “duck-curve” interval.

9.2 What explains the 2,097 proxy-positive hours?

All 2,097 proxy-positive hours are net-export hours and also contain positive non-VRE generation and Power-to-Heat consumption. Table 3 reports quantities only on those hours, so its columns are accounting components rather than mutually additive estimates of curtailment.

Table 2: Observed occurrence of the DK1 VRE-above-load proxy.

Sample	Positive hours	Valid hours	Positive-hour share
All complete hours	2,097	7,920	26.48%
Midday, 10:00–15:59	565	1,980	28.54%
July 2026	324	744	43.55%
August 2026 complete days	244	624	39.10%
Local hour 16:00	103	330	31.21%
Local hour 17:00	103	330	31.21%

Table 3: Accounting decomposition on the 2,097 proxy-positive hours.

Quantity	Aggregate value
VRE-above-load proxy, X^{proxy}	1,580,970 MWh
Net export, E	2,811,562 MWh
Non-VRE generation, C	1,230,592 MWh
Power-to-Heat already included in load, H	679,051 MWh
Export coverage, EC	100.00%
Non-VRE share of net export, CS	43.77%
Power-to-Heat counterfactual share, HS	30.05%
Maximum absolute hourly balance residual	1.6×10^{-5} MWh

The measured balance gives $2,811,562 - 1,230,592 = 1,580,970$ MWh up to numerical precision. Consequently, the original proxy is the portion of net exports left after accounting for non-VRE generation; it is not energy shown to be unused in DK1. If Power-to-Heat were removed only as an accounting counterfactual, the proxy on these hours would rise to 2,260,021 MWh. Observed Power-to-Heat therefore absorbs 679,051 MWh, or 30.05% of that counterfactual quantity, and must not be deducted a second time from gross consumption.

9.3 Gate-available proxy forecast results

The forecast evaluation contains 4,990 hours at each gate across 208 delivery days. The spring daylight-saving day, the incomplete final settlement day, and 24 weather-feature omissions on 24 June are not evaluated; the 7- and 14-day lag construction additionally leaves one local hour missing on 5 and 12 April. No missing value is filled. The minimum weather-availability margin before either decision gate is four hours.

Table 4: Gate-available component and proxy forecast performance.

Gate	Scope	Wind MAE	Solar MAE	Load MAE MWh	Proxy MAE	Proxy RMSE	Precision %	Recall
07:30	All hours	408.8	238.8	224.7	180.4	410.3	79.19	61.92
12:00	All hours	377.7	231.4	221.5	174.9	403.8	80.66	66.10
07:30	Midday	468.3	506.6	278.1	276.4	562.0	61.92	54.20
12:00	Midday	445.3	490.0	270.9	270.3	553.2	64.11	59.54

Moving from 07:30 to 12:00 reduces all-hour proxy MAE by 3.06% and RMSE by 1.59%; midday reductions are 2.20% and 1.57%, respectively. The mean absolute gate-to-gate proxy revision is 41.98 MWh, and the positive-hour classification changes in 243 of 4,990 common hours (4.87%). The later run therefore adds information, although the update is incremental rather than a regime change.

The forecasts are conservative in energy magnitude. Against 1,143,143 MWh of observed proxy energy, the 07:30 and 12:00 models predict 669,964 MWh (58.61%) and 724,365 MWh (63.37%), respectively. In the midday window the corresponding shares are 71.03% and 76.23%. The signal is sufficiently precise to define a soft proxy-aware reward or diagnostic, but its recall and energy underprediction do not support treating it as a hard estimate of all renewable-rich hours.

9.4 Strategy-level midday–evening outcomes

Table 5: Mean daily linkage outcomes on 210 common complete days.

Strategy	Midday charge (MWh/day)	SOC at 17:00 (MWh)	Evening reserve (MW h/day)	Evening delivery rate (%)	Recovery cost (EUR/day)
B2 FB0	6.173	6.867	18.444	95.64	482.7
B2 FB0 gate	6.071	6.850	18.444	95.64	473.9
B3 historical paired, $\chi = 0$	1.026	6.526	19.200	96.41	220.0
B3 historical paired, $\chi = 0.5$	1.110	6.633	18.718	96.05	216.2
B3 pooled Gaussian, $\chi = 0$	1.902	6.720	18.047	96.49	255.6
B3 pooled Gaussian, $\chi = 0.5$	2.382	6.868	17.645	96.43	259.2
X06 rebuild	2.668	6.879	18.628	95.95	322.9
X09 rebuild	1.988	6.767	18.269	96.82	234.1
X14 rebuild	0.509	5.979	19.618	95.64	214.7

The five-hour evening window has a physical maximum of $4 \text{ MW} \times 5 \text{ h} = 20 \text{ MW h/day}$. The observed commitments in Table 5 therefore occupy 88.2–98.1% of this power-duration envelope. Requested evening activation is only 8.4–8.6% of committed reserve energy. Additional midday energy cannot materially increase reserve capacity when the power commitment is already near 4 MW; its primary physical value is to raise the pre-peak SOC and protect delivery.

B2 performs the most midday charging, but its recovery purchases cost approximately EUR 474–483/day. The scenario-based methods reduce this to EUR 215–323/day. B2 recovers less energy than several stochastic methods but pays approximately EUR 54/MWh of recovered energy, compared with EUR 22–32/MWh for the scenario-based strategies. Its weakness is therefore recovery timing, not merely recovery volume.

X14 is the clearest reserve-oriented reconstruction: it commits 19.62 MW h/day in the evening but charges only 0.51 MWh/day at midday. X09 has the highest evening delivery rate, 96.82%, while retaining moderate midday charging and recovery cost. The pooled Gaussian strategy at $\chi = 0$ gives the highest mean daily total settlement value among the displayed methods, but its higher midday charging relative to historical pairing is accompanied by lower evening reserve and delivered activation. Increasing χ from zero to 0.5 raises midday charging and 17:00 SOC in the pooled Gaussian case, but does not improve the evening delivery rate or reserve commitment.

9.5 Diagnosis of the observed problem

On the matched decision days, the midday proxy contains 319,236 MWh. The largest overlap is 449.5 MWh for B2 FB0, so even the maximum observed proxy-absorption ratio is only

$$\max_m AR_m^{\text{proxy}} = \frac{449.5}{319,236} = 0.001408 = 0.1408\%. \quad (62)$$

For the scenario-based and published-method reconstructions, the corresponding range is only 0.0116–0.0697%. Among strategies, 34–44% of midday charging coincides with proxy-positive hours, compared with a proxy-positive midday-hour share of 31.27%. Prices therefore create weak indirect timing alignment, but the charging magnitude is far too small to represent material proxy absorption.

The observed linkage statistics are

$$\Gamma_S \in [0.33, 0.51], \quad \Gamma_R \in [-0.52, -0.26], \quad DR^{\text{eve}} \in [95.64\%, 96.82\%]. \quad (63)$$

The correlations are descriptive rather than causal. Their joint pattern is nevertheless informative: midday charging consistently raises the 17:00 SOC, but days with more midday charging do not receive more evening reserve. This is consistent with the frozen gate sequence and the near-saturation of reserve power, rather than with a failure of the battery to carry energy from midday to the evening.

Across the full day, 74–86% of undelivered activation is attributable to gross-power headroom, while only 14–26% is attributable to the SOC-energy envelope. Evening shortfall averages only 0.049–0.073 MWh/day. Thus the remaining delivery problem is mainly simultaneous use of battery power by the energy schedule and activation, not inadequate evening SOC.

Table 6: Evidence-based diagnosis of the five candidate explanations.

Candidate explanation	Evidence-based diagnosis
Insufficient midday charging	Primary gap. Only 0.51–2.67 MWh/day for the advanced methods and no more than 0.1408% proxy absorption for any method.
Excessive SOC reserved for aFRR	Ex-post slack, not the main bottleneck. SOC at 17:00 is 5.98–6.88 MWh, but the one-hour reserve requirement is 4.339 MWh and the energy constraint rarely binds.
Inadequate evening delivery	Secondary. Aggregate delivery is 95.64–96.82%; average shortfall is at most 0.073 MWh/day.
Excessive activation-recovery cost	Material for B2; reduced by stochastic scheduling. The cost is EUR 474–483/day for B2 versus EUR 215–323/day for scenario methods.
Midday charging and evening reserve are already coordinated	Physical link exists; decision link is incomplete. Midday charge raises SOC, but $\Gamma_R < 0$ and reserve is fixed at the earlier gate and already near its power ceiling.

10 Research claim and identified gap

The empirical claim of the present DK1 study is:

For a price-taking 4 MW/8 MWh battery participating in day-ahead energy and upward aFRR, probabilistic scheduling improves reserve value, activation delivery, and recovery timing, but does not by itself produce material charging aligned with the DK1 VRE-above-load proxy. Midday charging increases the energy available at 17:00, while the evening reserve quantity remains largely decoupled because it is fixed at 07:30 and already close to the battery power limit. The proxy-positive observations represent renewable-attributable net export pressure rather than measured curtailment. Gate-safe weather and load models provide a conservative signal before both decisions, but the frozen optimizer does not use it. The missing mechanism is therefore an explicit link between the 07:30 proxy forecast, the 12:00 update, charge timing, and the value of reliable evening upward-aFRR delivery.

This claim is narrower than a claim of actual curtailment reduction. It is useful to distinguish the proposed decision chain

$$\hat{X}_{d,h}^{\text{proxy},07:30} \longrightarrow r_{d,h}^{\text{up}}, S_{d,17}^{\text{target}} \longrightarrow \hat{X}_{d,h}^{\text{proxy},12:00} \longrightarrow P_t^{\text{ch}} \longrightarrow S_{17} \longrightarrow e_{17:21}^{\text{del}} \longrightarrow C^{\text{rec}}, \quad (64)$$

from the chain implemented in the current experiment. At present, $\hat{X}^{\text{proxy},g}$ is evaluated but does not enter u_d ; X^{proxy} enters only the ex-post diagnostic. The data show a positive charge-to-SOC link, but no positive charge-to-reserve link and almost no proxy overlap. A subsequent

model may therefore test a soft proxy-aware charge reward or SOC target at 12:00 and a forecast-informed energy budget at 07:30, but such a modification is not part of the present diagnostic run.

The numerical evidence is stored in the proxy-surplus, decomposition, and gate-forecast tables, the daily and summary midday–evening tables, and the nine strategy-specific dispatch traces. The experiment result manifest records the exact output names. The trace-to-daily aggregation error is at most 3.64×10^{-12} , confirming that the new ledger reproduces rather than changes the frozen settlement results.

11 Aggregation rules and admissible claims

Metrics are evaluated only on common intervals for which all required variables are observed. Missing values are not replaced by zeros, interpolated, or extrapolated. For multi-day or full-sample results, ratios are computed as the sum of their numerators divided by the sum of their denominators; daily ratios are not averaged without weights. If the denominator of Eq. (28) or Eq. (29) is zero, the corresponding daily metric is recorded as missing rather than zero.

Table 7: Permitted and prohibited interpretations.

Metric or evidence	Permitted interpretation	Prohibited interpretation
Eq. (9) with complete inputs	Curtailment rate under the stated system boundary	Replacing available generation with realized generation
Eq. (11)	Model-based reduction under a controlled system counterfactual	A price-taking DK1 battery changed observed market clearing
Eq. (26)	Simulated charging in the fixed midday window	Actual renewable energy utilized at midday
Eqs. (27)–(29)	Temporal alignment with the local VRE-above-load proxy	Every charged MWh came from VRE or reduced curtailment one-for-one
Eqs. (20)–(25)	Accounting decomposition of proxy-positive hours	Exports or Power-to-Heat are equivalent to observed curtailment
Eq. (38)	Model forecast of the reproducible ex-post proxy	Official TSO load forecast or physical-curtailment forecast
Eq. (31)	Simulated charging at a high VRE-to-load ratio	Curtailment necessarily occurred above the threshold
Eq. (32)	Simulated charging during negative-price intervals	Negative prices are equivalent to renewable curtailment

The primary terminology for the present study is “charging aligned with renewable-rich midday periods,” “potential renewable absorption,” and “midday-to-evening energy shifting.” The terms “actual renewable utilization rate” and “actual curtailment reduction” require unconstrained available VRE, observed curtailment or dispatch-down information, and a system-level counterfactual market-clearing model.

12 Relation to established formulations

Equations (7)–(10) follow the common system-dispatch distinction between available VRE and utilized VRE. The National Renewable Energy Laboratory (NREL) represents curtailment as the unused portion of otherwise available variable generation and evaluates storage within the system balance [1]. Its formulation for DC-coupled PV–battery systems further tracks curtailment recovered by storage charging within an explicit energy balance [2]. Peng et al. jointly model

generation, curtailment, transmission, and storage operation in SWITCH-China and constrain renewable-connected batteries to charge from co-located renewable generation [3]. These studies can evaluate system curtailment or renewable charging provenance because their models contain system balances, available generation, or source-coupling constraints that are absent from the present price-taking DK1 simulation.

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